

number			
8(a)	A description that makes reference to the following points: • (protein that) consists of 4 {subunits / (poly) peptides} (1) • has (four) {haem / non-protein / prosthetic} groups (1) • {iron ion / ferric iron / Fe²⁺} (attached to each subunit / haem) (1)	ACCEPT α and β, α and γ, δ and ε DO NOT ACCEPT α helix / β sheet IGNORE proteins, bonds DO NOT ACCEPT iron NB Has four haem groups that contain iron ion = 2 marks	(3)

Question number	Answer
*8(b)	Mean number of cells per dm^3 of blood • no correlation with size of mammal / some indication that as size of mammal decreases the number increases, with the exception of camels Mean mass of haemoglobin: • some indication that mass of haemoglobin per cell decreases with size of mammal, with the exception of camels / man • mammals with high numbers of red blood cells have lower mean mass of haemoglobin per cell / converse • total mass of haemoglobin decreases with size of mammal • because high number would partially compensate for less haemoglobin as the smaller mammal may require less energy • so respiration would be less • total mass of haemoglobin decreases with size of mammal Mean cell volume: • cell volume decreases with size of land mammals, except camels • possibly because their blood vessels are narrower • cell volume of camels is particularly large • possibly because their blood vessels are wider • as more heat needs to be lost in the hot temperatures Oxygen-carrying capacity: • oxygen carrying capacity of blood decreases with decrease in size of mammal • oxygen-carrying capacity decreases with total mass of haemoglobin • as the smaller mammal may require less energy • so respiration would be less Overall: • these three factors determine the oxygen-carrying capacity of the blood • but other factors would also affect oxygen-carrying capacity • e.g. affinity of haemoglobin for oxygen • beluga whales have relatively small cells but high mass of haemoglobin and high concentration enabling their high oxygen-carrying capacity • as swimming takes a lot of energy • as the water they live in is cold • and energy is needed to keep warm • therefore rate of respiration is high • and the animals have to surface to acquire their oxygen from the air

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple descriptions of data / correct comments not related to data / attempt but not accurate description of trends 1 mark = one description / comment 2 marks = two descriptions / comments
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Trends / lack of trends described with some explanation 3 marks = one correct trend described / an inaccurate trend explained 4 marks = one trend plus a simple explanation of data
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Explanations of data 5 marks = more than a simple explanation of a trend 6 marks = explanation that shows some linkage of data sets, that is discussed in relation to size of mammal

Question number	Answer	Additional guidance	Mark
8(c)(i)	• 26 (1) • 23 (1)	Bald answer of 23 = 2 marks Bald answer of 26 = 1 mark	(2)

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8(c)(ii)	A description that makes reference to the following points: - as partial pressure (of oxygen) drops from 12 to {1 / 2 / 3 / 4} (kPa) the percentage saturation of oxygen {remains at 80 to 100 % / remains fairly constant / drops slightly} (1) - partial pressure (of oxygen) drops sharply at low levels of oxygen (1) OR - percentage saturation (of oxygen) drops sharply below {1 / 2 / 3 / 4} (kPa) (1) - percentage saturation (of oxygen) {remains at 80 to 100 % / remains fairly constant / drops slightly} at high partial pressures (1)	NB Penalise reference to rate once ACCEPT converse throughout ACCEPT value within the range of 1 and 4 which lies on a grid line ACCEPT value within the range of 1 and 4 which lies on a grid line	(2)

number			
8(c)(iii)	An answer that makes reference to two of the following points: • because myoglobin has a higher affinity for oxygen (than haemoglobin) (1) • so that it will (only) release the oxygen when partial pressures are low (1) • because haemoglobin cannot supply enough oxygen to respiring muscle cells / so that myoglobin can supply <u>enough</u> oxygen to respiring muscle cells (1)	ACCEPT converse stronger {binding / association} with oxygen IGNORE easier / better ACCEPT converse ACCEPT muscle cells need <u>more</u> oxygen for respiration (than other cell types)	(2)